

6.100L Recitation 2 – September 16, 2022

Reminders:

- MQ 2 next Wednesday
- PS1 half way hand in due next Wednesday
- Finger exercises before each lecture

Lecture 2 Recap: Strings, Input/Output, Branching

Strings

- New data type – it is a sequence of characters
 - `my_string = "Hello world!"`
- They can be indexed and sliced:
 - `my_string[0]` # outputs "H"
 - `my_string[2]` # outputs "l"
 - `my_string[-1]` # outputs "!"
 - `my_string[-2]` # outputs "d"
 - `my_string[1:3]` # outputs "el"
- We can concatenate strings
 - `my_new_string = my_string + ' ' + my_string`

Input

- Done with the *input* command
- Anything the user inputs is read as a string object!
 - `x = input("Enter a string: ")` # what the user inputs is assigned to x as a string
- Can cast a user input as an integer
 - `x_as_int = int(input("enter and int: "))` # here x will be an integer

Output

- Done with the *print* command
 - `print(x)`
 - `print(("x = ", x) -> (comma concatenates with a space between)`
 - print statements are super useful for debugging! especially to see what is happening in loops

Branching

- Idea that we only want to execute certain blocks if specific conditions are satisfied
- We create a code structure to satisfy our requirement.

Example branching:

```
x = 2
if x == 3:
    print("x is 3!")
elif x == 2:
    print("x is 2!")
else:
    print("x is neither 2 or 3")
```

Lecture 3 Recap: Loops & Iteration Methods

Looping Mechanisms

- Loop over ranges of numbers
- Loop over elements of a string
- Main idea – want to repeat things multiple times → reuse code.

For loops

`for` loops have pre-specified range over which they run.

```
for i in range(x):  
    ▪ i goes from 0 to x-1  
for char in s:  
    ▪ char is string that takes on the value of each character in s
```

While loops

`while` loops have a condition that they check to determine if they should keep running. They run until the condition no longer evaluates to True.

```
counter = 0  
while counter < 3:  
    print(counter)  
    counter += 1
```

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6.100L Introduction to CS and Programming Using Python

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